higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N5

1 April 2025

This marking guideline consists of 7 pages.

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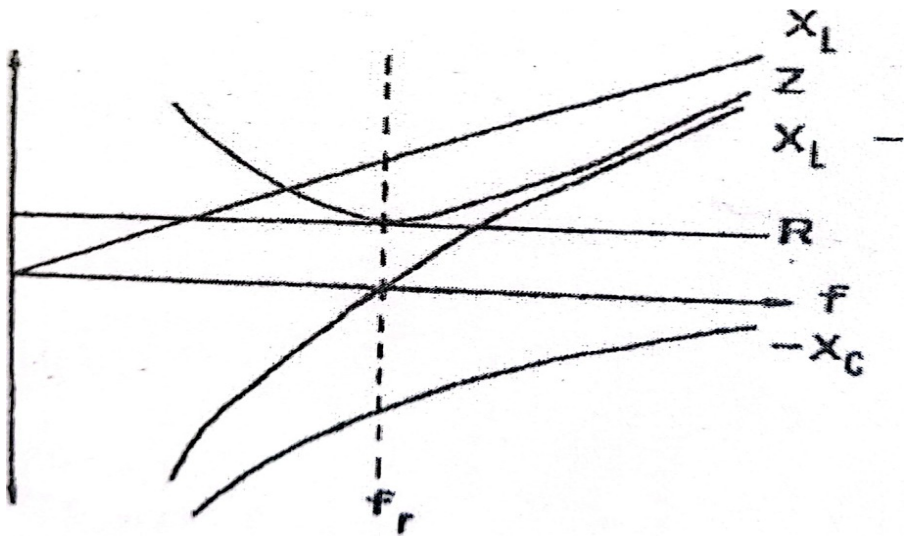
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NATED Question Papers	**N1-N6** Mathematics NATED	**N1-N4** Engineering Science NATED	**N1-N6** Industrial Electronics NATED	**N4-N6** Electrotechnics NATED	**N4-N6** Mechanotechnics NATED
N5-N6 Power Machines NATED	**N4-N6** Quantity Surveying NATED	**N5-N6** Strength of Materials & Structures NATED	**N1-N3** Engineering Drawing NATED	**N3** Eletro-technology NATED	**N3** Mechano-technology NATED
N4-N6 Building Administration NATED	**N4-N6** Building& Structural Surveying NATED	**N4-N6** Building & Structural Construction NATED	**N1-N3** Building Science NATED	**N1-N3** Building Drawing NATED	More…
N4-N6 Computer Practice NATED	**N4-N6** Entrepreneurship and Business Management NATED	**N4** Management Communication NATED	**N4-N6** Communication NATED	**N4-N6** Public Administration NATED	**N4-N6** Financial Accounting NATED
N4-N6 Computerised Financial Systems NATED	**N5-N6** Sales Management NATED	**N4-N6** Office Practice NATED	**N5-N6** Public Finance NATED	**N4-N6** Personnel Management NATED	**N5-N6** Municipal Administration NATED
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QUESTION 1: ALTERNATING CURRENT THEORY

1.1 It is a receiver's ability to discriminate between signals having different frequencies.✓ (1)

1.2



Allocate half mark for any six correctly labelled curves. (3)

1.3 1.3.1 $I_A = \frac{V_T}{Z_A}$

$\text{¿} \frac{250 \angle 0°}{15 - j15{,}915}$

$\text{¿} \frac{250 \angle 0°}{21{,}87 \angle -46{,}7°}$

$\text{¿} 11{,}43 \angle 46{,}7° A$

$I_B = I_T - I_A$

$\text{¿} 16 \angle 0° - 11{,}43 \angle 46{,}7°$

$\text{¿} 16 + j0 - (7{,}839 + j8{,}318)$

$\text{¿} 8{,}161 - j8{,}318$

$\text{¿} 11.653 \angle -45{,}54° A$ (6)

1.3.2 $Z_B = \frac{V_T}{I_B}$

$¿ \frac{250 \angle 0°}{11.653 \angle -45,54°}$

$¿ 21,453 \angle 45.54° \Omega$

$Z_B = 15,024 + j\,15,313\,\Omega$

$R = 15,024\,\Omega$ and

$L = \frac{X_L}{2\pi f} = \frac{15,313}{2\pi \times 50} = 48\,mH$ (4)

[15]

QUESTION 2: POWER SUPPLIES

2.1 2.1.1 To give more smoothing on the pulsating output DC voltage.✓ (1)

2.1.2 To regulate voltage during varying of load conditions.✓ (1)

2.2 2.2.1 $V'_{DC} = \frac{R_L}{R_L + R_S} V_{DC}$

$¿ \frac{1000}{1000+10} \times 14$

$¿ 13,86\,V$ (3)

2.2.2 $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times 100\,x\,10^{-6}} = 15,915\,\Omega$

$V'_{R(RMS)} = \frac{X_C}{\sqrt{R^2 + X_C^2}} X_C$

$¿ \frac{15,915}{\sqrt{10^2 + 15,915^2}} \times 0,8$

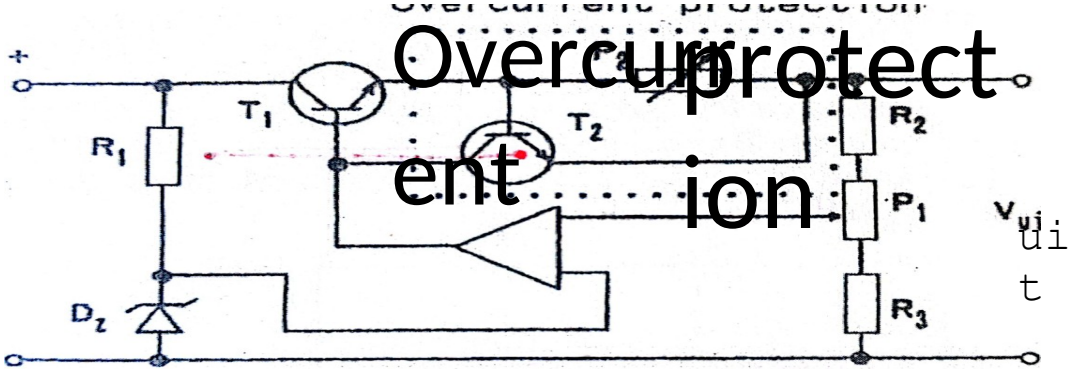
$¿ O\,,677\,V$

$\%r = \frac{V'_{R(RMS)}}{V'_{DC}} \times 100$

$¿ \frac{0,677}{13,86} \times 100$

$¿ 4,9\,\%$ (5)

2.3



Allocate HALF mark for any correctly labelled components. (5)

[15]

QUESTION 3: AMPLIFIERS

3.1 3.1.1 Reasonable gain.✓
3.2.1 Maintenance.✓
3.2.1 Collector current.✓

(3 × 1) (3)

3.2 3.2.1 $V_E = I_C R_E$

$\therefore 450 \times 10\, x\, 10^{-3}$✓

$\therefore 4{,}5\,\Omega$

$V_E = \frac{V_{CC}}{10}$

$V_{CC} = 10 \times 4{,}5$

$\therefore 45\,V$ (4)

3.2.2 $C_E = \frac{10}{2\pi f R_E}$

$f = \frac{10}{2\pi C_E R_E}$

$f = \frac{10}{2\pi \times 450 \times 15\, x\, 10^{-6}}$

$\therefore 235\,Hz$ (3)

3.3 3.3.1 B✓
3.3.2 D✓
3.3.3 E✓
3.3.4 C✓
3.3.5 A✓

(5 × 1) (5)

[15]

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QUESTION 4: INTEGRATED CIRCUIT AND TRANSDUCERS

4.1
- Use earthing/grounding wristband.✓
- Connect all test equipment to earth.✓
- All unutilised input must be connected to V_{DD} or V_{ss}.✓
- Switch off the power before you work on the circuit✓ (4)

4.2
$$Resolution = \frac{V\ drop\ across\ two\ adjacent\ turns}{Total\ V\ drop}$$

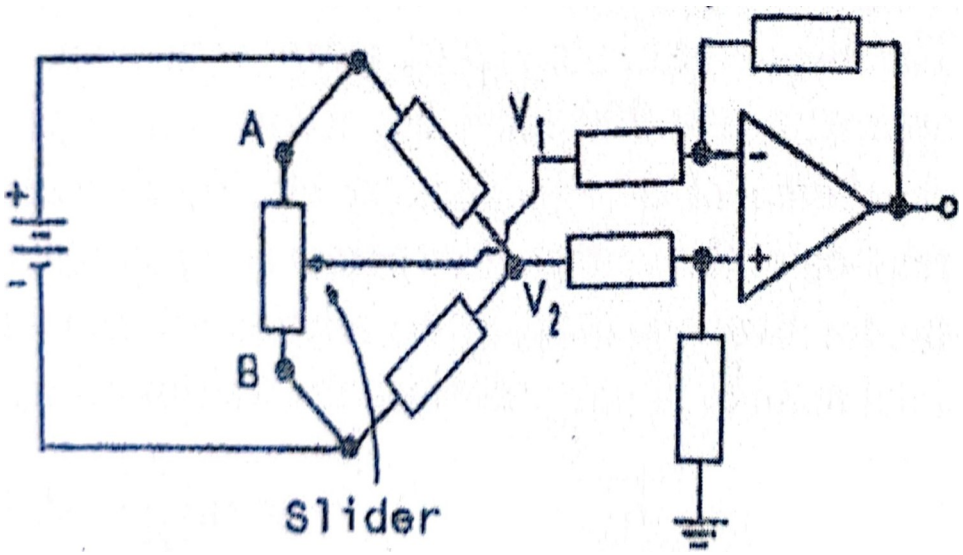
$$0{,}03 = \frac{15\ x\ 10^{-3}}{Total\ V\ drop}$$

$$Total\ V\ drop = 500\ mV$$

(3)

4.3
- Measuring temperature.✓
- Control temperature.✓ (2)

4.4



Allocate half mark for correctly labelled components. (6)
[15]

QUESTION 5: ELECTRONIC PHASE CONTROL

5.1
- Reducing of anode to cathode current.✓
- Line commutation - phase control.✓
- Force commutation - DC supplied circuit.✓ (3)

5.2 As soon as the load deviates,✓ the feedback signal will differ from the reference input.✓ This difference is called the error signal✓ and is used by the control circuit to bring the load back to normal.✓ As the load returns to the normal condition,✓ the error signal will become smaller until it finally equals zero.✓ This way the load is automatically kept constant without any human interaction.✓ (7)
[10]

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QUESTION 6 : MEASURING INSTRUMENT

6.1
- The stair slope voltmeter.✓
- The double slope voltmeter.✓
- The voltage to frequency converter.✓
- The successive approximation voltmeter.✓
- The frequency counter. (Any 4 × 1) (4)

6.2 6.2.1 Switch actuator – sends signals to reference voltage.✓

6.2.2 Flip-flop – transfers the pulse to the switch.✓

6.2.3 Counter – counts the signals as soon as the gate closes.✓

6.2.4 Gate – accepts the signal from the comparator.✓

6.2.5 Display unit – displays the results on the screen.✓

6.2.6 Clock pulse generator – starts to count as soon as the gate opens and stops when the gate closes.✓

(6 × 1) (6)
[10]

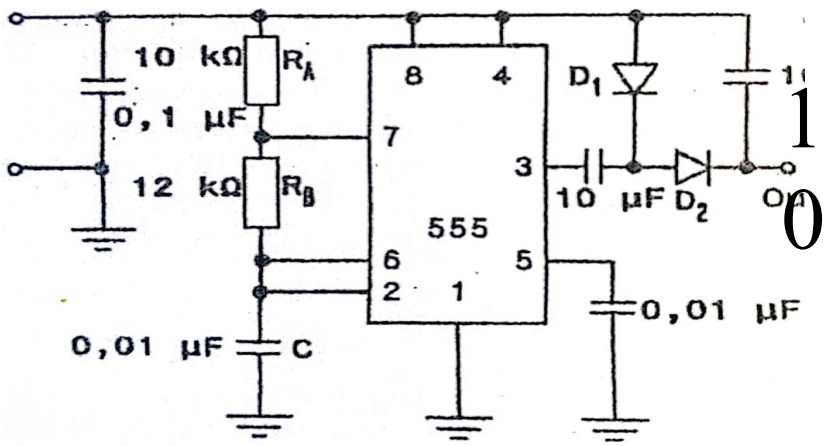
QUESTION 7: OSCILLATORS

7.1
- Power supply.✓
- Positive feedback.✓
- Resonant frequency and feedback signals is in phase. (Any 2 × 1) (2)

7.2 7.2.1 False✓
7.2.2 False✓
7.2.3 True✓

(3 × 1) (3)

7.3



Allocate HALF mark for correct and labelled components. (5)
[10]

QUESTION 8: LIQUID CRYSTAL DISPLAY

8.1

Passive matrix	Active matrix
• It has a grid of conductors with pixel at each intersection.✓	• It has a transistor at each pixel intersection, which requires less current to adjust pixel brightness.✓
• A current is sent across two conductors in the grid to control a pixel's light.✓	• The need for less current allows pixels to switch ON and OFF more rapidly.✓
• Some models use dual scanning to improve the speed.✓	• The rapid action results in a faster screen refresh rate.✓

(6)

8.2
- TV sets.✓
- Computer monitors.✓
- Aircraft cockpit display.✓
- Indoor and outdoor signs.✓
- Small screens in projectors.

(Any 4 × 1) (4)

[10]

TOTAL: 100

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